

MDM End Semester Examination Preparation Notes

A. Problem Statement (5 Marks)

Research Problem

Traditional corporate finance research evaluates CEOs primarily through financial metrics and assumes that they act as rational decision-makers. However, non-verbal cues such as CEO photo prominence in annual reports, facial width-to-height ratio (fWHR), and vocal characteristics during earnings calls may reveal important aspects of managerial personality.

Most previous studies have focused on only one type of cue (facial or vocal) and have largely been conducted using Western firm samples.

Purpose of the Study

To examine whether multiple CEO non-verbal signals jointly influence:

- Firm Performance (ROA, ROE)
- Financial Risk (Stock Return Volatility, Leverage)

among large Indian listed firms (BSE 100) over a ten-year period (FY2015–16 to FY2024–25).

Theoretical Foundation

- Upper Echelons Theory
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B. Method: Operationalization (Empirical) (10 Marks)

1. Data

- Sample: 50 BSE-listed Indian firms
- Time Period: FY2015–16 to FY2024–25
- Total Observations: Approximately 500 firm-year observations
- Data Sources:
 - Annual Reports
 - Secondary Financial Databases
 - CEO Photographs from Annual Reports
 - Earnings Call Audio from YouTube

2. Variables

A. Dependent Variables (DV)

Firm Performance

- Return on Assets (ROA)
- Return on Equity (ROE)

Financial Risk

- Stock Return Volatility
- Leverage Ratio

Market Value

- Tobin's Q
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B. Independent Variables (IV)

Photo-Based Variables

- Photo Presence
- Photo Size
- Photo Location
- Photo Prominence Score

Facial Variables

- Facial Width-to-Height Ratio (fWHR)
- Facial Symmetry
- Likeability Score

Vocal Variables

- Pitch (F0 Mean)
- Speech Rate
- Pause Rate
- Voice Energy (RMS)
- Transcript Sentiment

These variables were combined into a:

Voice Confidence Index (using PCA)

C. Control Variables (CV)

- Firm Size (Log Revenue)
 - Revenue Growth
 - CEO Age
 - CEO Tenure
 - CEO Gender
 - Board Size
 - Board Independence
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3. Research Method

Research Design

- Quantitative Panel Data Design

Tools Used

Photo Analysis

- MediaPipe Face Mesh
- MTCNN
- OpenCV
- DeepFace

Audio Analysis

- Librosa
- openSMILE

Statistical Technique

- Random Effects GLS Panel Regression
 - Industry Fixed Effects
 - Year Fixed Effects
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D. Practical / Social Implications (5 Marks)

1. Boards and Nomination Committees

CEO selection may be influenced by facial appearance and vocal characteristics. Therefore, boards should adopt structured and objective evaluation procedures to minimize bias.

2. Investors and Analysts

Markets react not only to financial data but also to behavioral signals. Non-verbal CEO characteristics can serve as supplementary indicators of leadership quality and firm risk.

3. Corporate Communication

CEO photographs and earnings-call presentations influence stakeholder perceptions. Firms should carefully manage these signals as part of investor relations strategies.

E. Contributions (10 Marks)

Contribution 1

First Indian study to jointly examine:

- Photo Prominence
- Facial Features (fWHR)
- Voice Confidence

within a single research framework.

Contribution 2

Provides unique evidence from an emerging market (India) using a ten-year panel dataset.

Contribution 3

Demonstrates that CEO behavioral traits are linked to:

- Accounting Performance (ROA, ROE)
- Financial Risk (Stock Volatility)

thereby connecting behavioral finance with accounting outcomes.

F. Key Learnings (10 Marks)

Learning 1

Accounting ratios such as ROA and ROE are influenced by leadership behavior and personality traits, not merely financial decisions.

Learning 2

Annual reports contain important non-financial signals. CEO photo placement and prominence can provide information about managerial identity and behavior.

Learning 3

Panel data analysis provides stronger and more reliable evidence than a single-year cross-sectional study because it captures firm behavior over time.

Quick Revision (1 Minute Before Exam)

Purpose

Study impact of CEO non-verbal cues on firm performance and risk.

Data

- 50 Firms
- 10 Years
- 500 Observations

DV

- ROA
- ROE
- Volatility
- Leverage
- Tobin's Q

IV

- Photo Prominence
- fWHR
- Voice Confidence

CV

- Firm Size
- Growth
- CEO Age
- Tenure
- Gender
- Board Characteristics

Method

- Panel Data
- GLS Regression
- Fixed Effects

Remember

- 3 Implications
- 3 Contributions
- 3 Key Learnings